**Name:** Team Timbuktu **Date:** 2/23/25

## *Element G: Construction of a testable prototype*

The final prototype of the toy enrichment hanger for the eland underwent substantial refinement throughout its development process, with each design iteration bringing us closer to the real design. Initially, our construction faced delays due to challenges in getting suitable materials, which hindered our team's progress. However, once we successfully gathered the necessary supplies, all the group members helped in assembling the prototype using two 2x4s, two small L-brackets, eye hooks, and cotton twine. While this initial design showed promise, it became clear that several adjustments were required to ensure the strength and stability necessary for its intended use. This realization prompted us to engage in multiple iterations of the design, each addressing specific weaknesses identified in the initial prototype.

The construction of the prototype emphasized durable materials, specifically utilizing 4x4s and 2x4s to ensure it could support the weight of the toys, which could be as heavy as 100 pounds. Recognizing the limitations of cotton twine, which stretched excessively under load and compromised the prototype's integrity, we quickly sought a more reliable alternative. The decision to replace the twine with rock-climbing rope marked a critical turning point in our design process. Rope provided the necessary strength to withstand the weight of the toys without deforming or risking structural failure. Additionally, we took measures to enhance the overall stability of the hanger by extending the base with a splint that incorporated an extra 2x4 length, thus bringing the design closer to the required scale and ensuring it met all specifications. To further bolster its structural integrity, we added two 1x4s at the joint between the top support beam and the vertical posts, effectively acting as a T-bracket that mitigated bending under any load.

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To ensure our final prototype met the desired performance attributes, we executed rigorous testing and modeling. This included simulating various weight loads up to the maximum of 100 pounds to rigorously verify both its stability and strength. We conducted calculations on stress and strain using Garrett, one of our group members, to act as the eland's toy. This testing phase enabled us to collect quantitative data reflecting the performance of our design, ensuring the hanger could withstand the anticipated forces while remaining functional and safe for the Eland. Additionally, the iterative approach we adopted allowed for continuous adjustments based on testing outcomes, enabling us to refine measurements and optimize the prototype for maximum efficiency, durability, and safety.

While our prototype's physical attributes can be quantitatively tested, we must also consider several non-testable qualities. For instance, the enrichment value of the toys and how the Eland interacts with them cannot be directly measured without observing their behavior over time. Understanding factors like the eland's interest level in specific toys and their ability to engage playfully will necessitate direct observation in a real-world environment, potentially leading us to implement behavioral studies to evaluate the enrichment solution's success. Furthermore, we need to reflect on the mental impact that enrichment tools have on the Eland, which is essential for promoting mind stimulation and well-being. This multifaceted approach emphasizes that, while physical construction and durability are paramount, so too are considerations for animal welfare and engagement, which ultimately define the success of the enrichment solution we are creating. By addressing both testable and non-testable attributes comprehensively, we aim to develop a toy enrichment hanger that not only functions effectively but also meaningfully enriches the lives of elands in their habitats, contributing to their overall well-being and happiness.